FALCON AUTOTECH

BLUE DART

FALCON's Response to BLUEDART for RFQ of Cross Belt Sorter_Gurgaon

August 27, 2024

www.falconautotech.com

Kind Attention –

Mr. Manoj C Gupta

Mr. Vivek Sharma

M/s Bluedart

Offer Ref: F24-00379-02; Date 27.08.2024.

Subject – Techno -Commercial Offer of CBS-8K_Gurgaon

We are pleased to submit our Techno-Commercial Offer to Bluedart in response to your **RFQ for Sorter, Gurgaon.**

We would like to highlight the fact that Falcon Autotech, as a prime contractor for this project, is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative. From technical point of view, the system is designed at 10,000 PPH and ensures simple operations and movement within the facility.

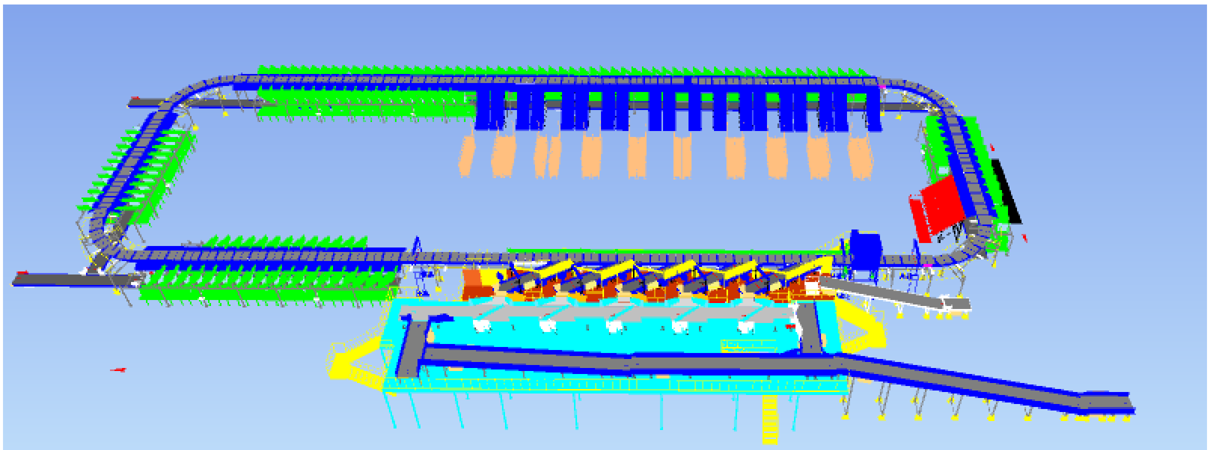
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Rishabh Jain

VP- Sales

Response to RFQ for CBS_ BLUEDART, Gurgaon



Proposal Reference – F24-00379-02

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Contents

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	ARB	Actuated Roller Balls
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this RFQ of Sorter Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.
As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the Sorter system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.
Our solution is based on the following key characteristics:

- **One sorter, based on a cross belt technology, offers an operational throughput of 8000 pph.**
- **Infeed conveyor system.**
- **5 Nos of Feedline**
- **19 PTL Chutes, 150 Direct Bagging Chute, 4 rejection chute, and 5 VDS chute.**

A tailor-made and simple layout, specifically designed to BLUEDART The proposed layout, is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our last Teams workshop meeting.

- Simple operational conditions due to one single linear cross belt sorter.
- Easy maintenance: optimized number of conveyors and concentrated inducts area.

1. **Falcon's reliable and fully CE compliant Parcel sortation systems**

These systems are globally being used by most innovative brands such as Aramex, Amazon, Asendia, Fastway, Delhivery and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the BLUEDART Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from the completion of installation & commissioning.

3. **Commitment to ensuring the quality of the delivery and its schedule.**

Falcon proposes a ramp-up approach, with a soft go-live by End of January 2025.

3. Handled Parcel Spectrum

As per the parcel spectrum data provided in the RFQ documents, Falcon has studied and analysed the parcel spectrum in detailed.

3.1 Parcel size loadable on the sorter

Falcon's heavy duty cross belt sorter has a capability to handle the below mentioned parcel sizes and weight.

Specification	Unit	Value
Max Length	mm	500
Max Width	mm	500
Max Height	mm	500
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	50
Min Height	mm	5
Min Weight	Kg	0.05
Dimension Accuracy	mm	+- 10
Weighing Accuracy	Kg	+/- 60 grams or 0.5% of the Weight of Product (whichever is higher)

Note- Dimension will only be captured for parcels > 20 mm in height .

3.2 Parcels to be loaded on Sorter shall have the following characteristics:

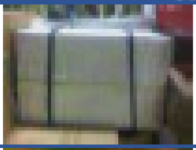
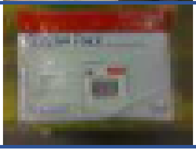
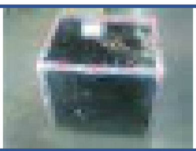
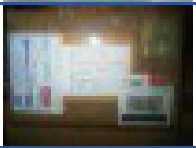
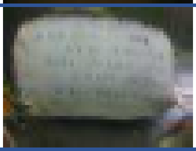
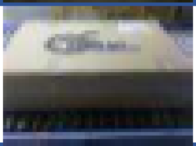
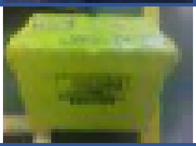
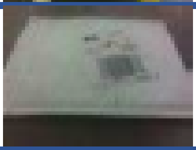
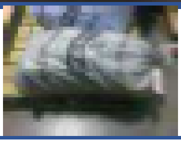
1. Centre of Gravity of item must not move during conveyance or sorting.
2. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
3. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
4. Plastic ropes shall be perfectly adherent to the surface of the package.
5. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
6. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
7. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
8. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
9. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
10. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

3.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

3.4 Pictures of Sortable Parcels

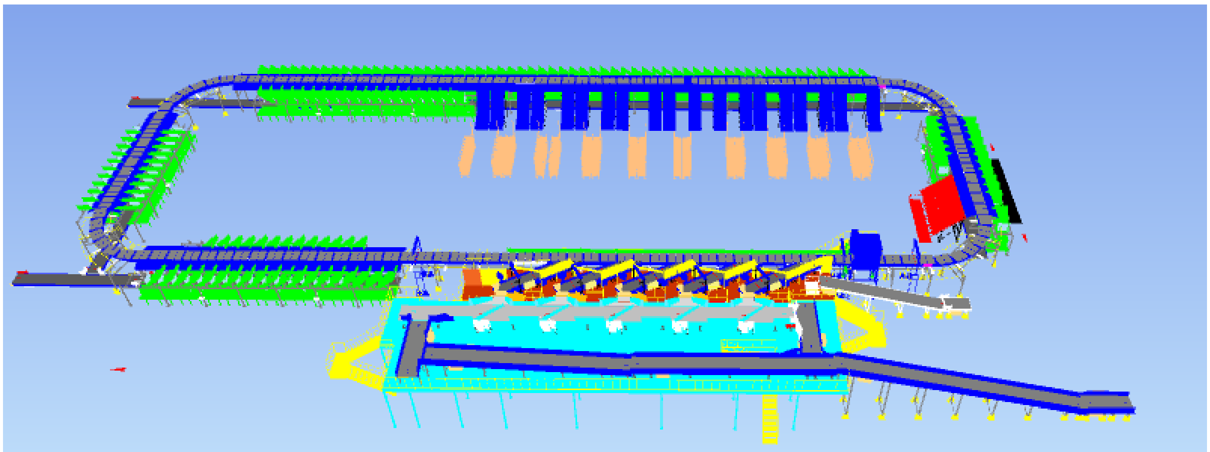
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

4. Proposed System Description

4.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Linear CBS for sorting shipments into destination as per specifications shared by BLUEDART.

4.2 Summary of the System



The complete parcel processing system includes, in accordance with the specifications provided by **BLUEDART:**

Process Flow:

- Shipments will be loaded on the Bulk Loading Conveyor placed on ground level
- Infeed Conveyor will move the shipments to the VDS Conveyor
- On VDS Conveyor, Arm VDS will distribute the shipment and sort them on the VDS Chute
- From the VDS Chute , Operator will pickup the parcels and placed it on the induct line.
- Operators orients and places the shipments on the Induct lines equipped with sensors
- Feedlines induct the parcels on the Sorter automatically onto to the Cross-Belt Sorter which then sorts the parcels into respective chute using data from BLUEDART's Sorting decision system. The Dimensions and image of the parcel is also captured on the Sorter.

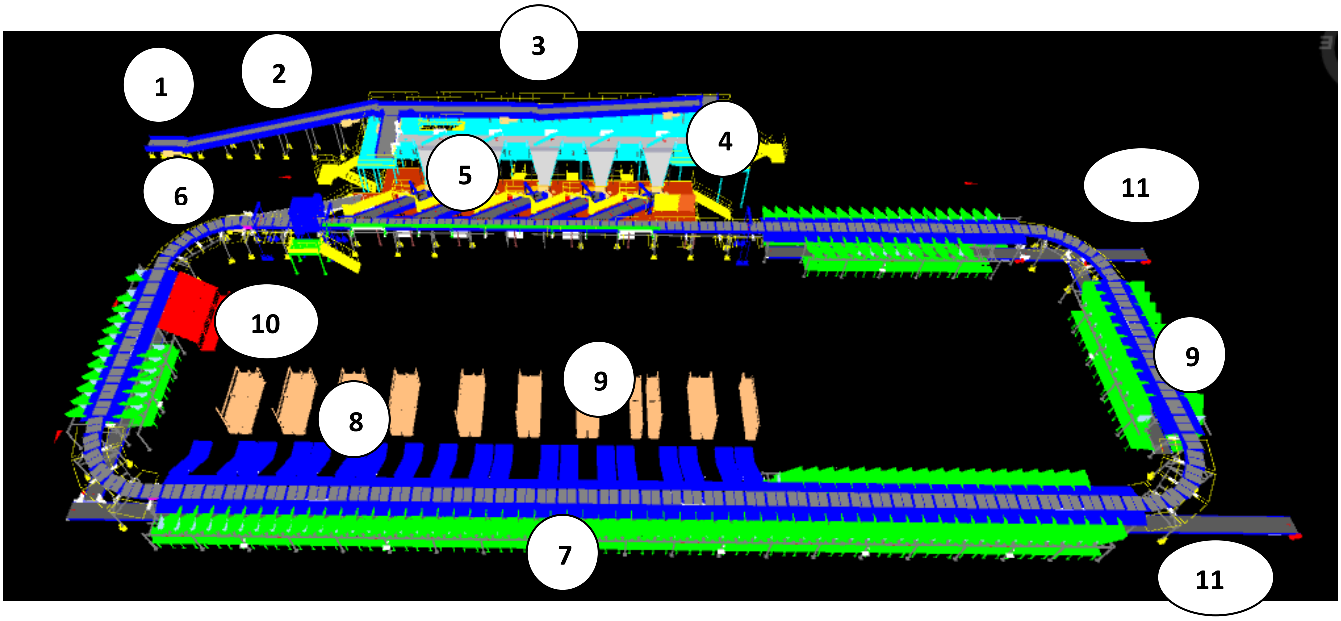
- The sorted shipments are discharged into PTL Chutes and Direct Bagging Chute.
- Secondary Sorting happens at PTL Chutes with help of Put to Light Racks and Bags are prepared. Associates scans the Shipments and drops them into the Bags guided by Light Modules on Racks.
- Bag created on the respective chutes will be dropped on bagging conveyors by the operator.
- Bagging Conveyor will transfer the bags to the bag collection point .

4.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 1.5 m/s.
3. Low occupancy of floor space in the building
4. Narrow discharge centers for increased number of splits in limited space
5. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

5. Layout View



Legend

1 – Bulk Loading Conveyor

2 – Infeed Conveyor

3 – VDS Loop Conveyor

4 – VDS Arm & VDS Chute

5- Feedline

6- Sorter

7- Direct Bagging Chute

8- PTL Chutes

9- PTL Frames

10- Rejection Chute

11- Bagging Conveyor

6. Proposed System Capacity Calculations

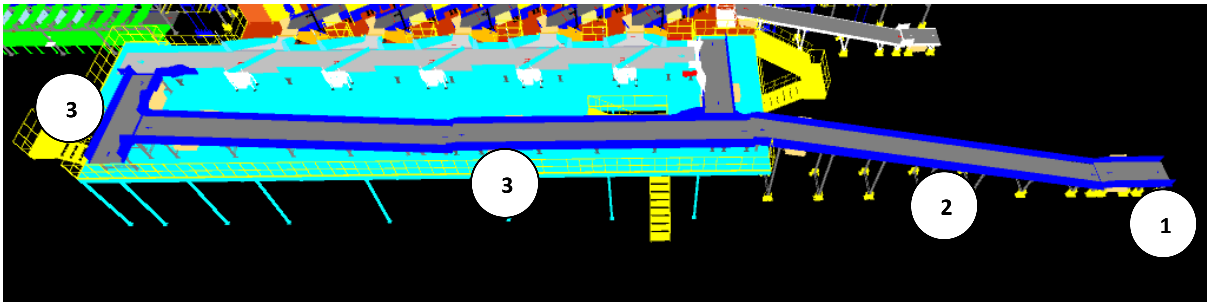
6.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on BLUEDART's RFP requirements.

SPECIFICATION	VALUE
Sorter Type	**Loop CBS**
Speed	**2.1m/s**
Pitch	**1175 mm**
Carrier per hour	**6,400 CPH**
Belts per hour	**12,800 BPH**
Sorter Designed Capacity	**12,800 PPH**
Feedline Designed Capacity	**2,000 PPH**
No of Inducts Planned	**5 Nos**
Total Induct Designed Capacity	**10,000 PPH**
Total Induct Operational TPH	**8,000 PPH**

7. System description

7.1 Infeed System



Legend

1- Bulk Loading Conveyor
2- Inclined Conveyor
3- VDS Loop Conveyor

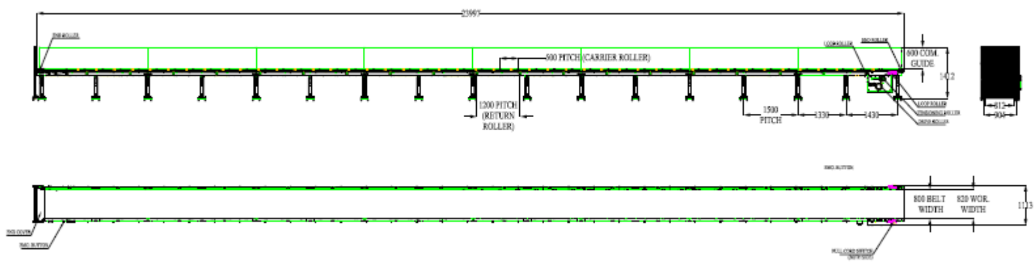
7.1.1 Straight & Inclined Powered Belt Conveyor

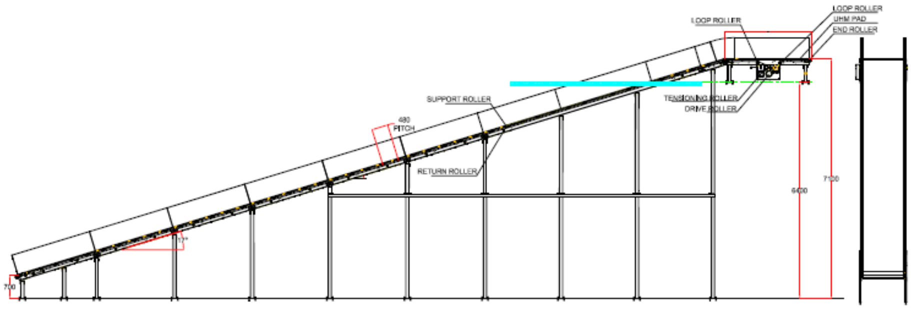
Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



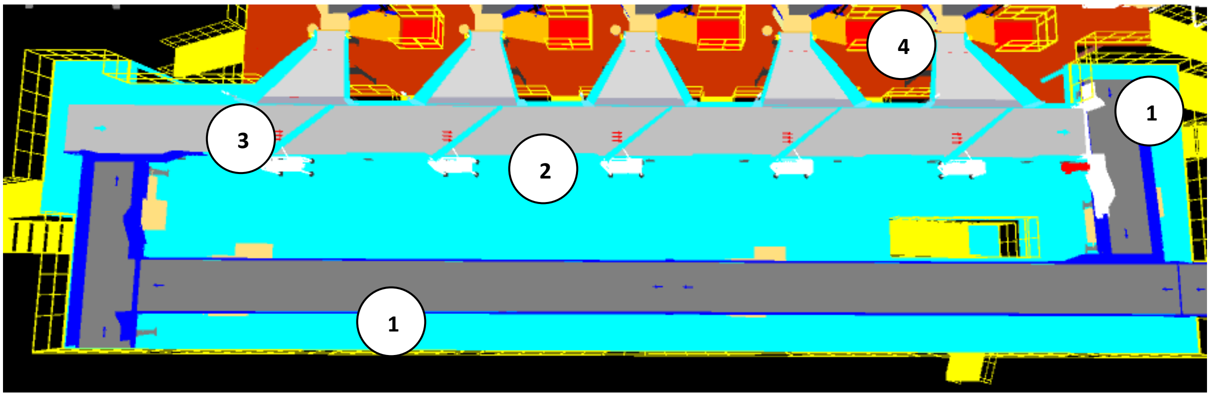


Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000/1200/1400
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord/Lenze
Drive	make	Lenze/Siemens/Omron/Allen Bradly/Mistubishi

7.2 VDS System:

Volume distribution system is designed to manage and distribute shipments efficiently along the feedlines. Volume distribution System has below key 18component:

- VDS Loop Conveyor
- VDS Modular Conveyor
- VDS Arm
- VDS Chute

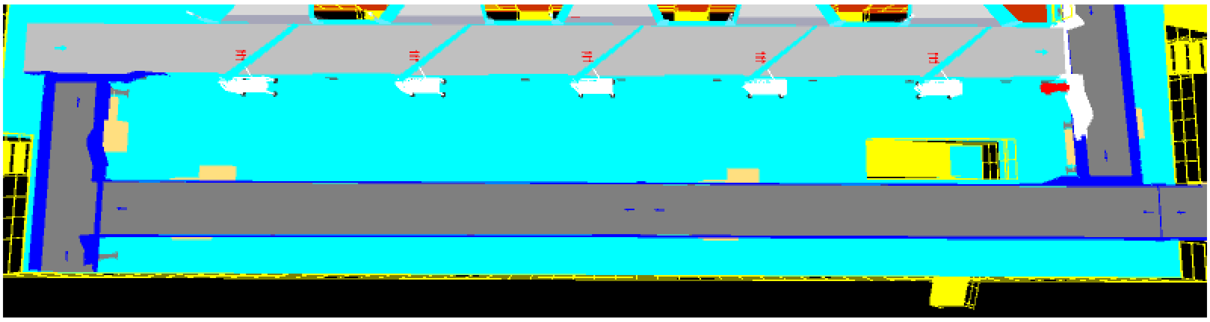


Legend

1. VDS Loop Conveyor
2. VDS Modular Conveyor
3. VDS Arm
4. VDS Chute

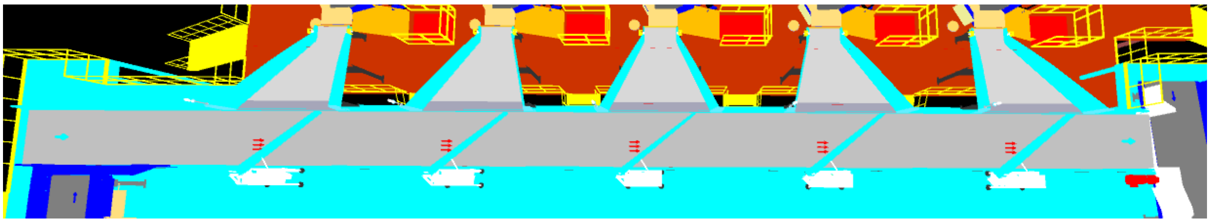
7.2.1 VDS Loop Conveyor:

Bulk materials will enter in to the VDS Loop through Infeed Highway Conveyors. Then the material will get distributed in to different VDS Chute maintaining load balance. The remaining material gets circulated in the VDS loop till the VDS Chute not be available for sorting .



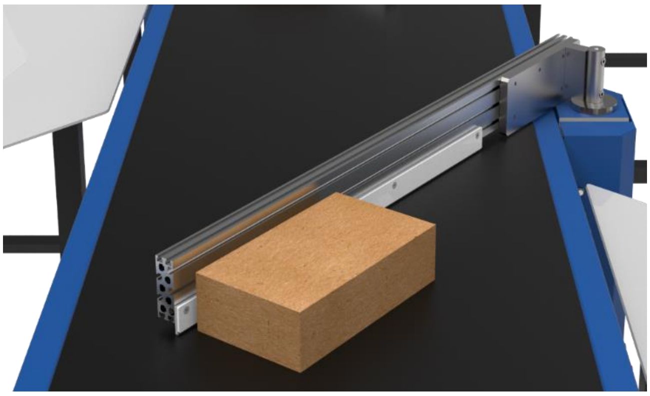
7.2.2 VDS Modular Conveyor

In this solution we have designed modular conveyor as VDS Conveyor .



7.2.3 VDS Arm

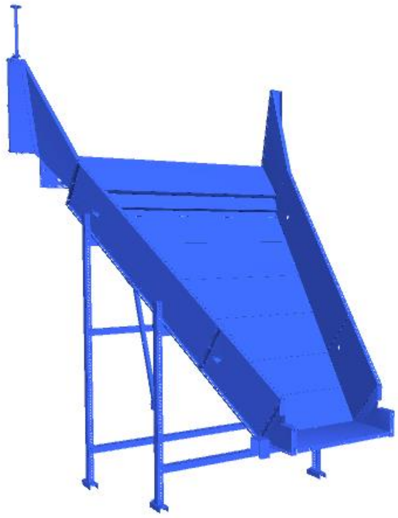
Linear Side Arm diverter System of Falcon AutoTech is an extremely cost-effective Sortation system designed for low to medium Volume Centres. Linear side arm diverter's simple and modular design gives it advantage of low cost, space saving, and modularity of increasing sorting points in future. This is one of the earliest diverter systems of Falcon AutoTech.



Specification	3.0 ARM XL
Property	Value
Actuation method	Electric induction motor
Belt type	Flat top modular
Arm length (mm)	2000

7.2.4 VDS Chutes

A VDS chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level (at an operator level). It utilizes base metal to guide and support the flow of items along the chute & collect the shipments within its area.



On each VDS Chute below compnents has been planned :

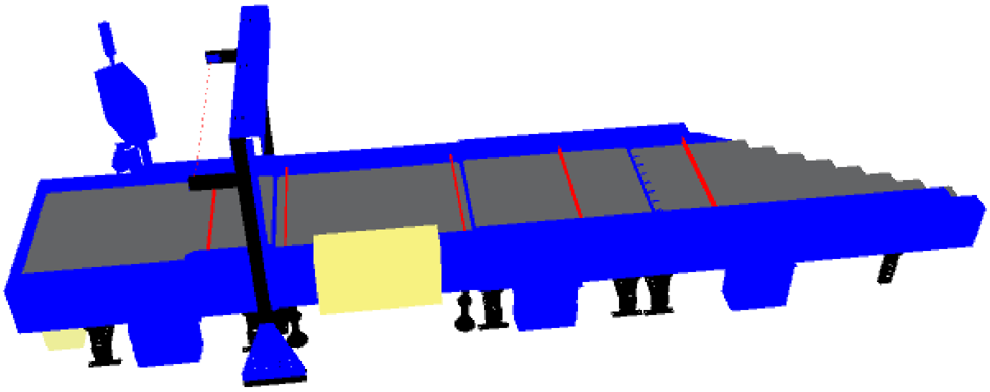
1. Chute Full Sensor – 2 Nos
2. Three Colour Tower Light – 1 No

7.3 Conveyor BOQ

S No.	Name	ANGLE	EL_1	EL_2	Conveyor Length (m)	Conveyor width (mm)	Set
1	PVC Belt Conveyor	0	5050	5050	4.32	1210	1
2	PVC Belt Conveyor	0	4500	4500	12.48	1210	1
3	PVC Belt Conveyor	0	800	800	3	1210	1
4	PVC Belt Conveyor	14	600	4700	16.8	1210	1
5	PVC Belt Conveyor	4.5	4300	5300	12.6	1210	1
6	PVC Belt Conveyor	5.1	4350	4750	4.32	1210	1
7	PVC Belt Conveyor	0	800	800	1.7	1010	1
8	PVC Belt Conveyor	15	698	3080	9.153	1010	1
10	Bagging Conveyor	-0.3	300	400	17.76	1010	1
11	Bagging Conveyor	-0.3	300	400	17.28	1010	1
12	Bagging Conveyor	-0.3	300	400	18.72	1010	1
13	Bagging Conveyor	-0.3	300	400	18.72	1010	1
14	Bagging Conveyor	-0.6	300	400	9.121	1010	1
15	Bagging Conveyor	-0.3	300	400	17.76	1010	1
16	Bagging Conveyor	-0.3	300	400	17.76	1010	1
17	Bagging Conveyor	-0.4	300	400	15.84	1010	1

7.4 Feedlines

Loop CBS has a total 5 Feedlines with a Designed IPP of 2000 PPH.



1. Each Feedline has four Conveyor Modules-

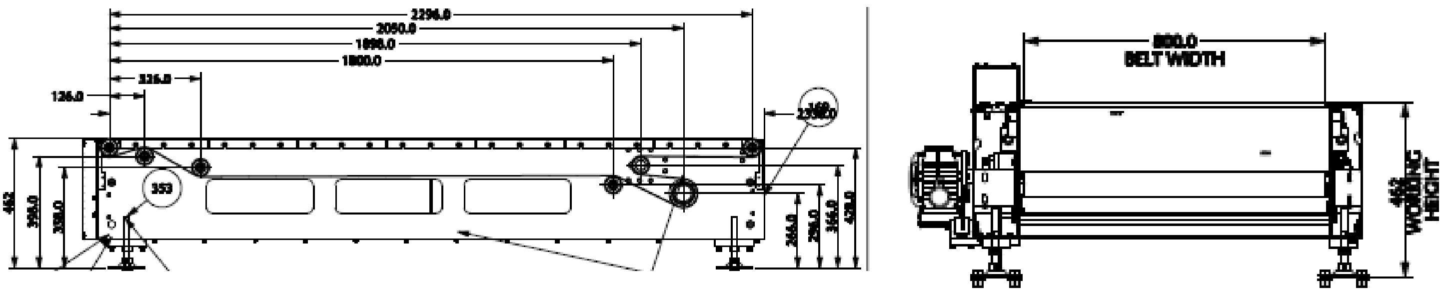
    - Loading Conveyor- 1 No.
    - Weighing Conveyor- 1 No.
    - Buffer conveyor- 1 No.
    - Intelligent Merge- 1 No.

2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.

3. Over the feedline, shipments are evenly spaced and smoothly transferred to the main Loop.

7.4.1 Loading Conveyor

Loading conveyor is a type of conveyor system used to align & Release the products for induction onto CBS. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.



7.4.2 Weighing Conveyor

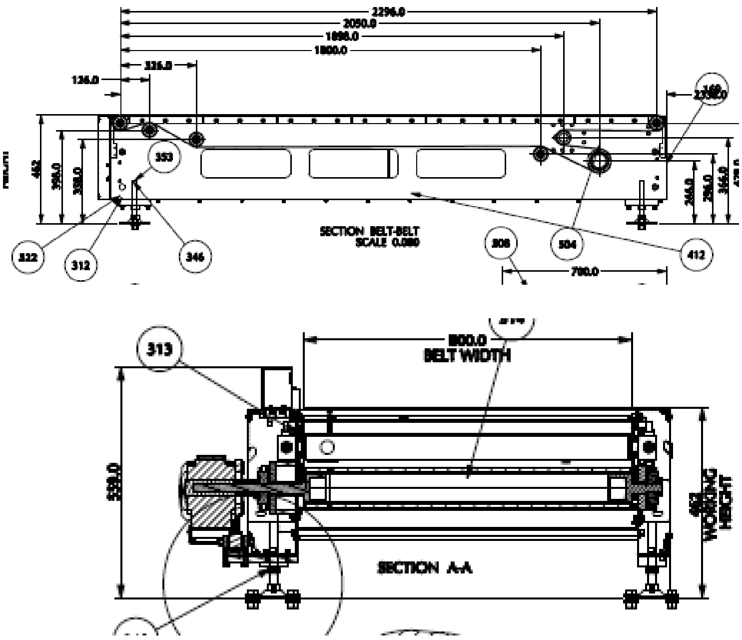
A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

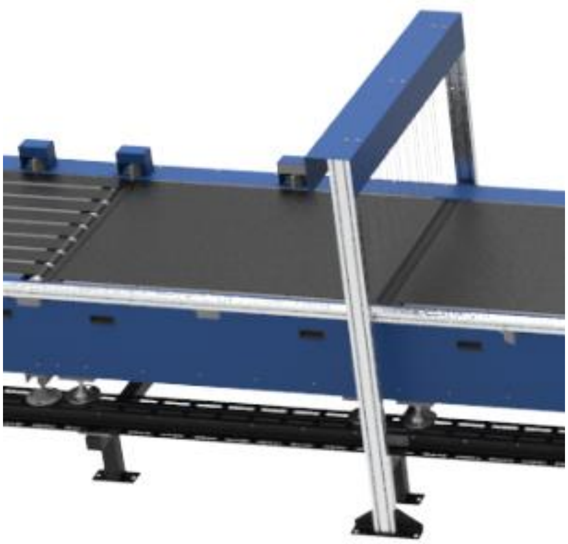
Make- Bizerba/ Mettler Toledo

7.4.3 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.

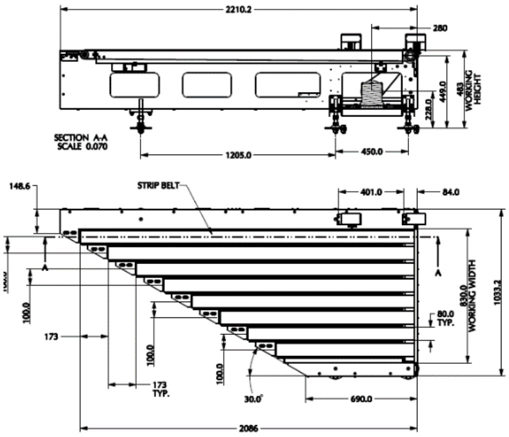




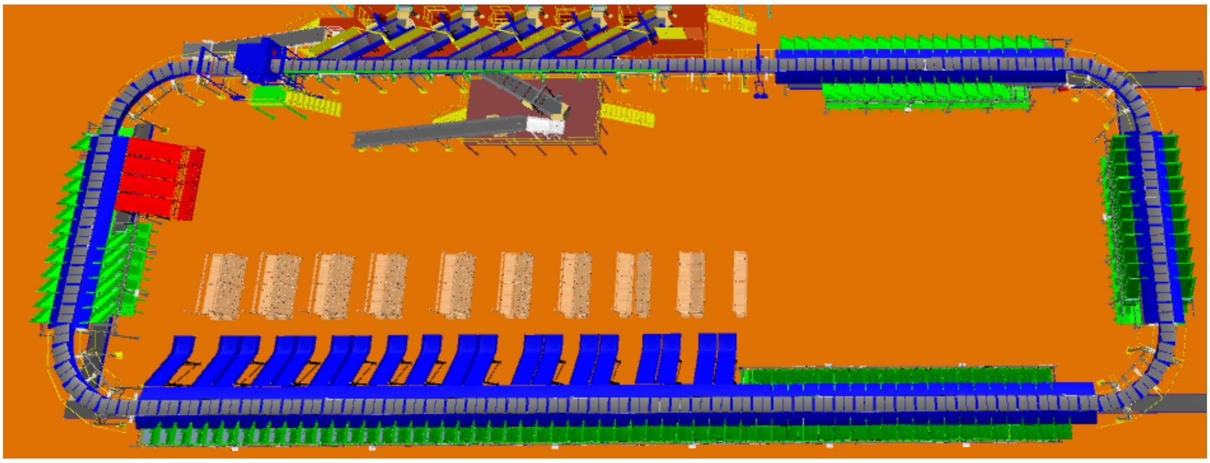
7.4.4 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



7.5 Sorter



1. The proposed CBS Sorter will be installed on ground. The Sorter runs at 3000 mm from ground level .

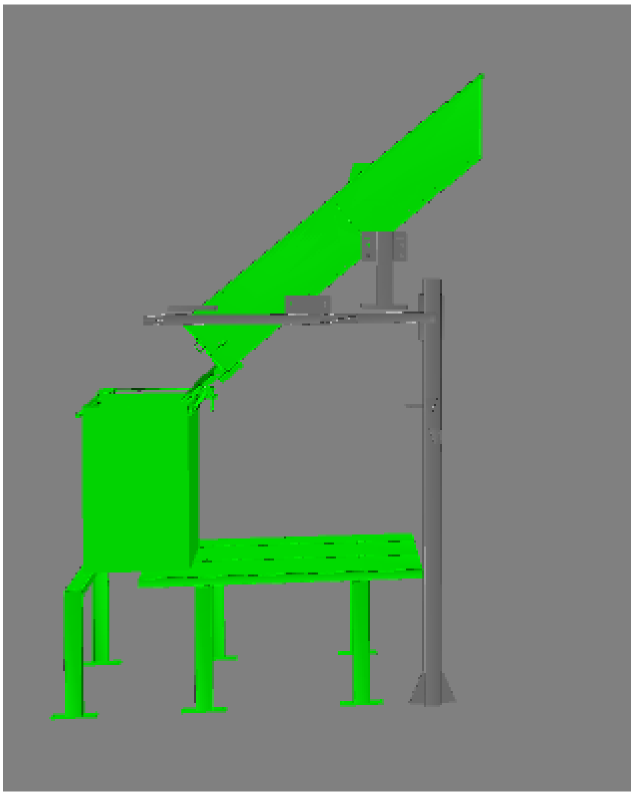
2. We have provided loop with dual Belt Carriers with 1175 mm pitch and Belt Size of 1000 x 800mm.

3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

7.6 Sorting Output

7.6.1 Direct Bagging Chute

Direct Bagging Chute is planned to sort the parcels directly in to the bags .

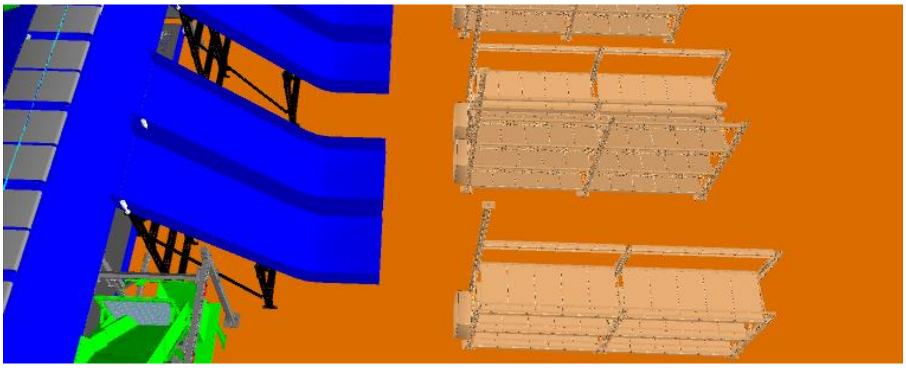


Each chute is equipped with below component

Chute IOs	Description
Single Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Tower Lamp	To indicate the operator action on a chute
Push Button	To Suspend/Resume Sorting into a particular Chute

7.6.2 PTL Chute

These are collection type chutes where the parcels are collected into chutes.The Operators collects the parcels from chute and will do secondary sorting by means of PTL Racks.

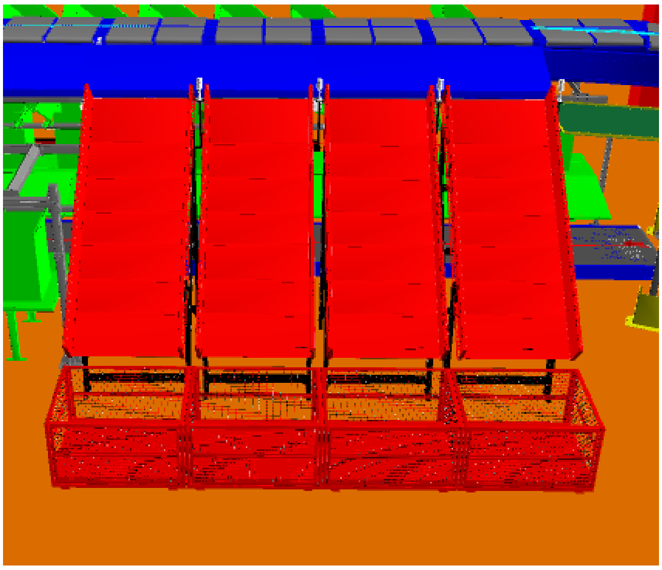


PTL Racks & Modules

- PTL Racks with PTL light modules has been provided for secondary sorting of parcels that have been dropped in the PTL chutes.
- Rack Configuration: 3 Rows, 8 Columns, total of 24 modules/rack in one chute.

- Operators will pick up a parcel from the PTL chute and scan the parcel barcode by handheld scanner.
- The light of the block in the PTL rack which has been mapped to the destination location of the parcel will glow.
- Operator will place the parcel in the bag framed in that module and acknowledge by pressing the confirmation button on the module.
- This process will be repeated for all parcels falling in that chute bin.
- Once a bag is full, the operator will take out the bag paste specific tag on bag seal and place it on the bag take-away conveyor.
- The new bag will be placed in the empty module.

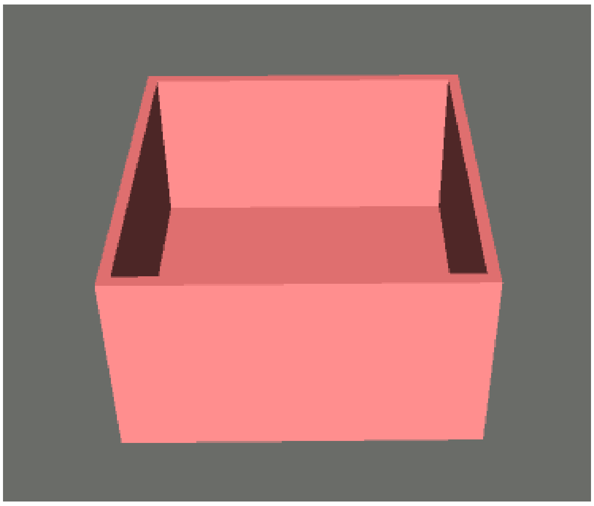
7.6.3 Rejection Chute



System is installed with 4 No rejection chutes Once the parcel got rejected, it will stage on the Collection Bins for the manual process.

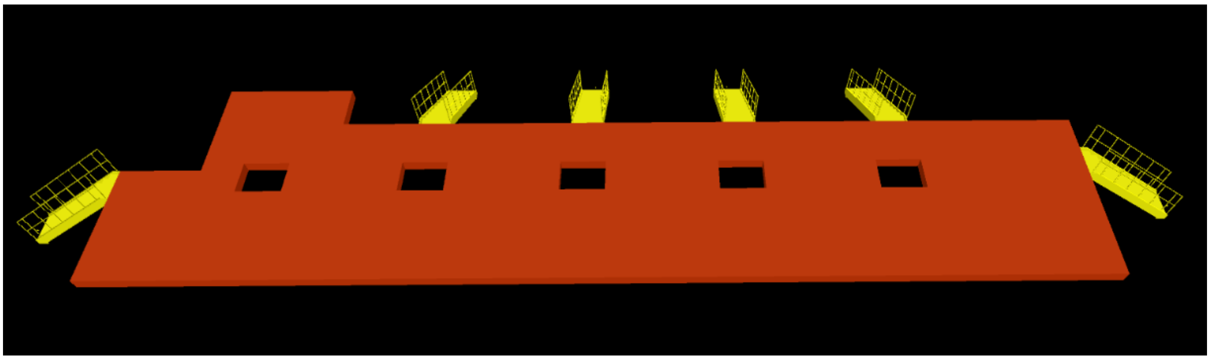
7.6.4 Collection Bin

As per Bluedart recommendation Spring Loaded Bin is planned on rejection chute and non-conveyable chute to collect the parcels.

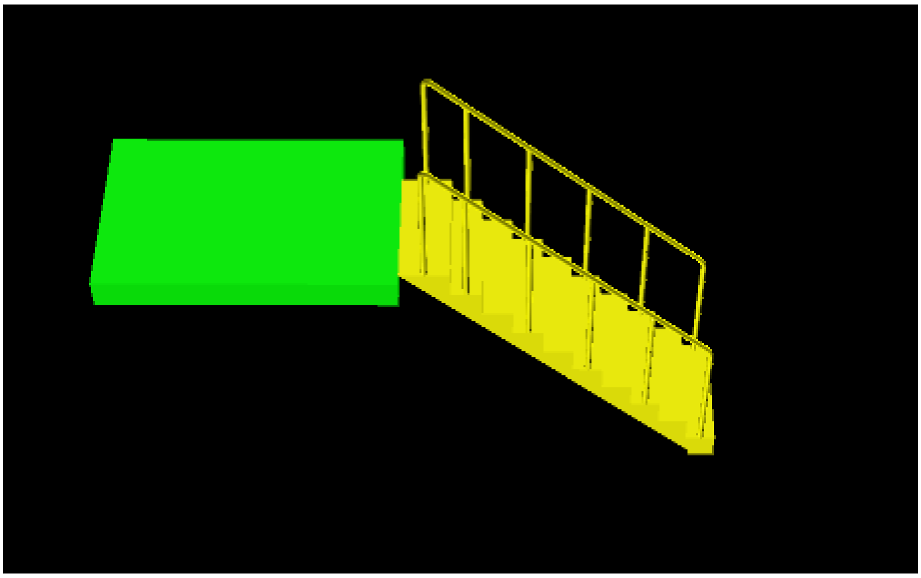


7.7 Mezzanine & Staircase

Mezzanine & Staircase for VDS System



Mezzanine & Staircase for Induct System



Platform for VMS Tunnel

8. Proposed System Technical Details

8.1 Mechanical equipment

Pos.	Qty.	Description	Value
FM1	1	**Infeed System** • Powered Belt Conveyors	 ~209 Metres (16 Conveyors).
FM2	5	**Feedlines** Consists of • Loading Conveyor. • Weighing System • Dynamic Spacing System • Angle merge • 1-D Hand Scanners for Manual Scanning	 1 Set 1 Set 1 Set 1 Set 1 No.
FM3	1	**Falcon Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch Including: • Standard Sorter Supports • Product Centring System • Camera based Barcode reader. • Dimension & Top Side Scanning System • E-Stops	 3000 mm ~164 m ~2 m/s Linear Motor 1175 mm
FM4	1	**Chutes** • Direct Bagging • 1-d Handscanner for Direct Bagging CHute • PTL Chutes • Rejection Chute • VDS Chute • Irregular Chute • Spring Loaded Bin	 150 Nos 16 Nos 19 Nos 4 Nos 5 Nos 5 Nos 10 Nos
FM5	1	**PTL** • PTL Racks for Chute • PTL Modules • 1-d Hand scanner for PTL Racks • PTL Control Box	 456 Nos 456 Nos 19 Nos 19 Nos
FM6	1	**Steel Structure** • VDS System Mezzanine • Induct System Mezzanine • Platform for VMS Tunnel • Stairs	 ~203 sq metres ~ 131 sq metres ~ 3 sq metre 10 Nos

8.2 Electrical Equipment

Electricals			
FE1	**1**	**Consists of** Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

8.3 IT Equipment

Electricals			
FI1	**1**	**Consists of** Server Rack including IT components required for server	Included

9. Description of Components of Equipment

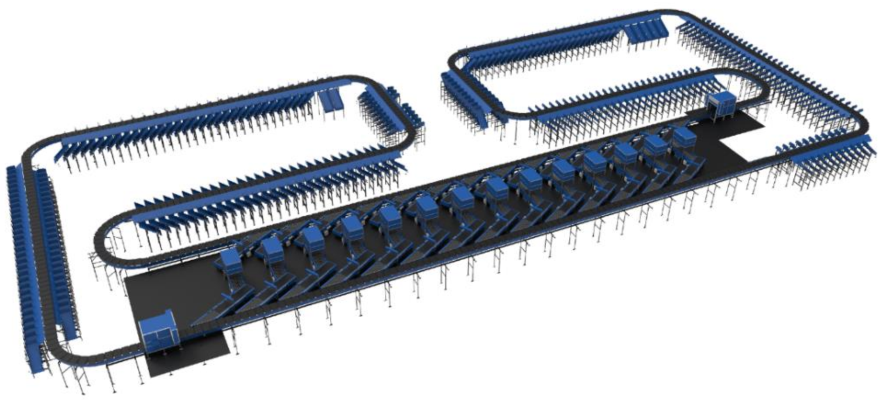
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

9.1 Cross Belt Sorter -

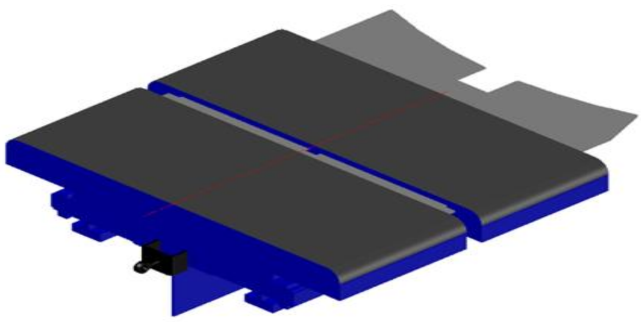
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



9.2 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners "and non-sortable parcel recirculation.



9.3 Servo Roller

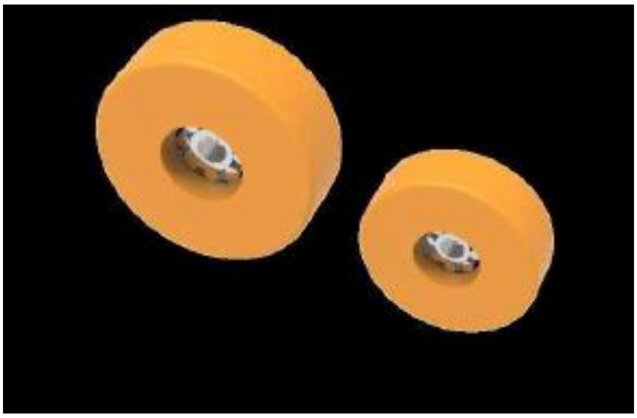
High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.

9.4 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.

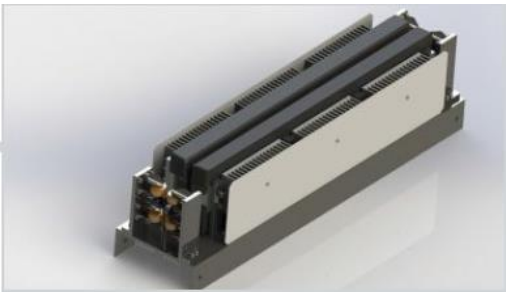
9.5 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle.



9.6 Non-contact based linear motor drive

No Contact based Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.

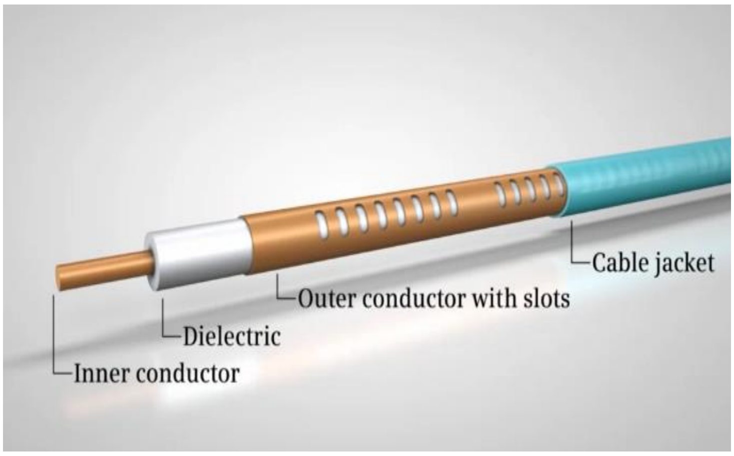


9.7 Power Transmission

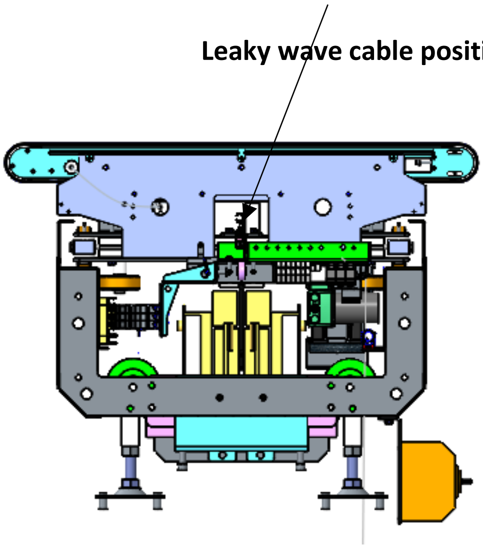
Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.

9.8 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.

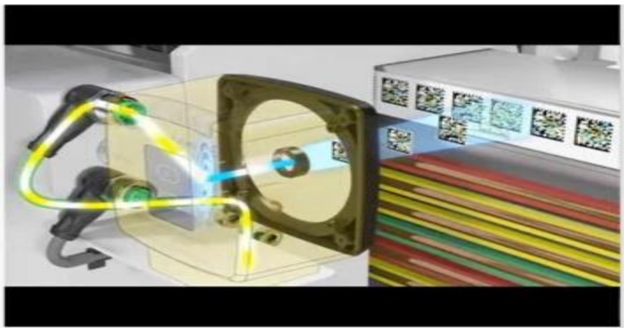


Leaky wave cable position



9.9 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



9.10 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length Per Deck	~164 m
Sorter Actuation Technology	Electric
Chutes	
Direct Bagging Chute	150
PTL Chutes	19
Rejection Chute	4
VDS Chute	5
Irregular Chute	5
Track Material to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during Detailed Design
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

9.11 Scanning & Sensing on Main Sorter Loop

Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.

Specification	UOM	Remark
Manufacturer Name	Name	SICK
Barcode Type	Type	1D&2D
Module Size	Mils	>=10 for 1D & >=25 for 2D
Handled Product Sizes	Mm	Min: 100 x 50x 5 Max: 500 x 500 x 500
Number of Codes	Nos	1
Location	Side	Top
Orientation	Type	Omnidirectional
Color of bars	Color	Black
Under Foil	yes/no	No

9.12 Parcel Positioning System.

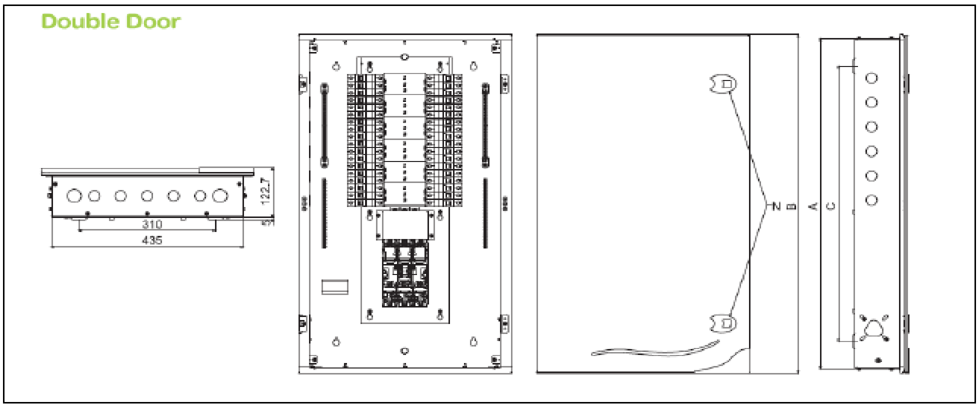
Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).

10. Electrical System

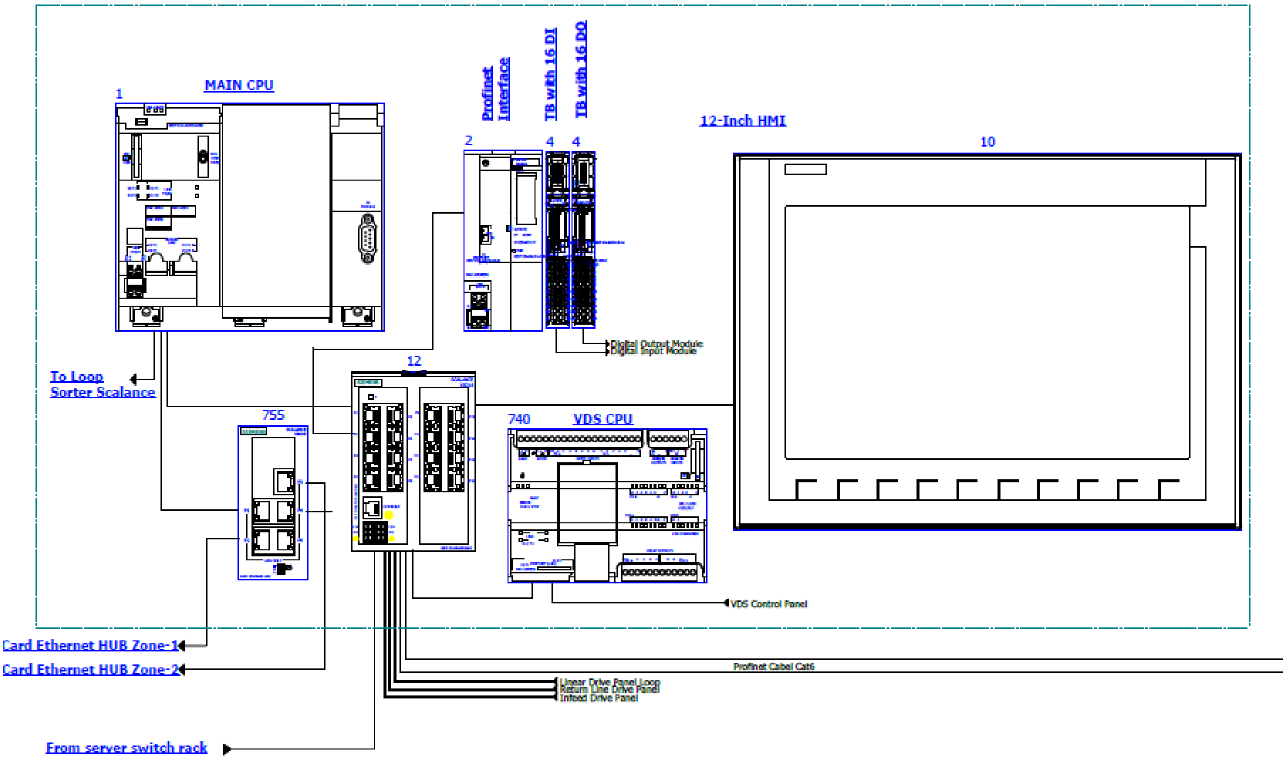
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

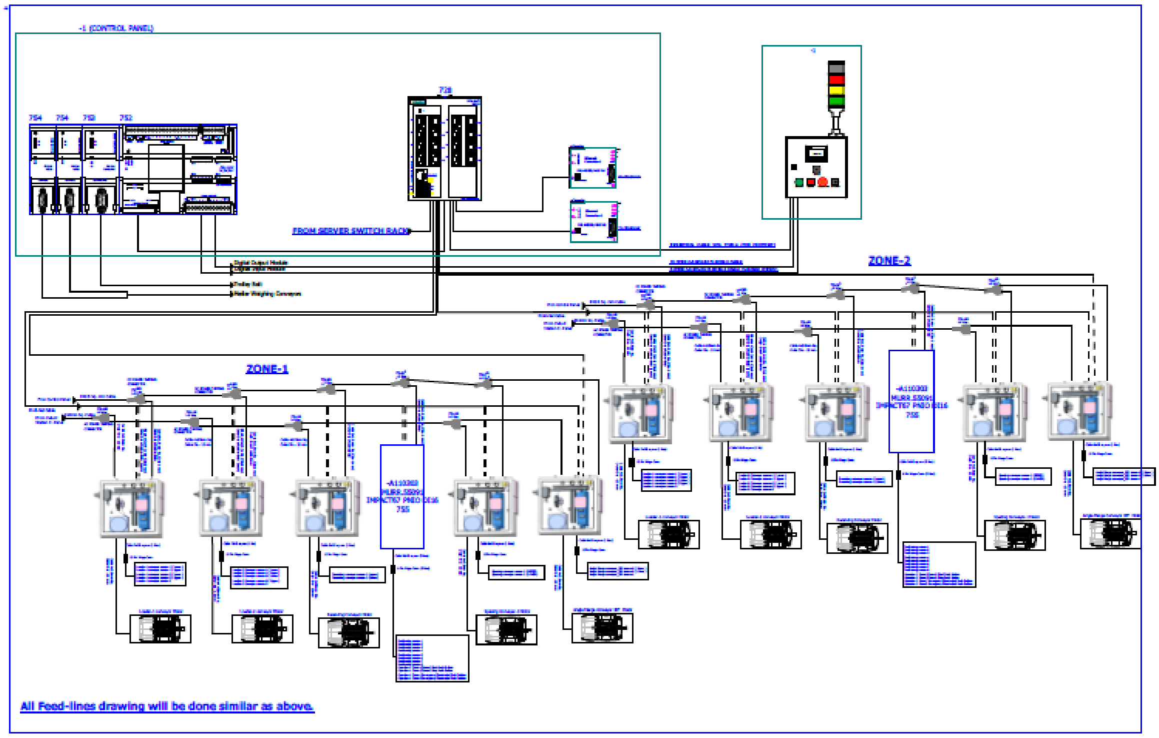
10.1 Reference Picture of Power Distribution Panel



10.2 Main Control Panel (Reference)



10.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.
All motors will have appropriate IP ratings

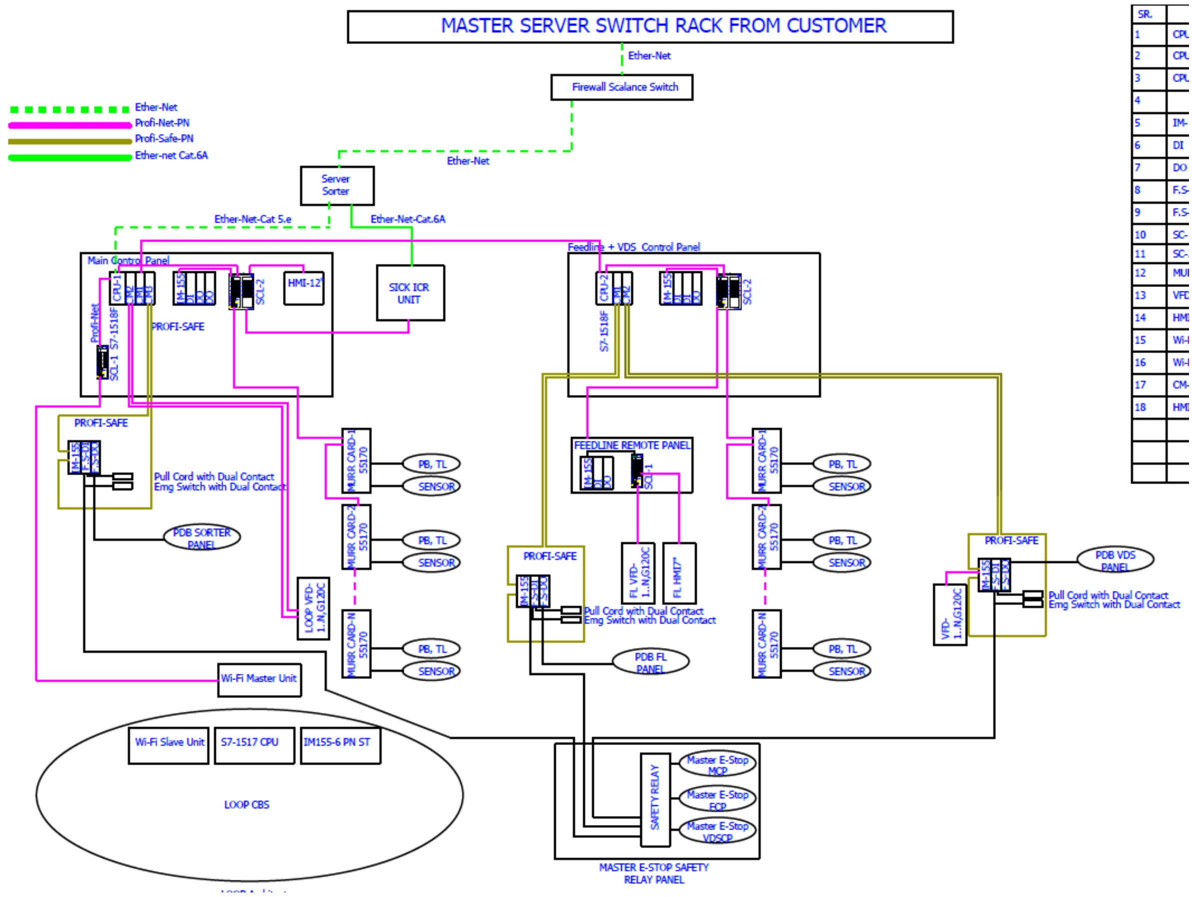
Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

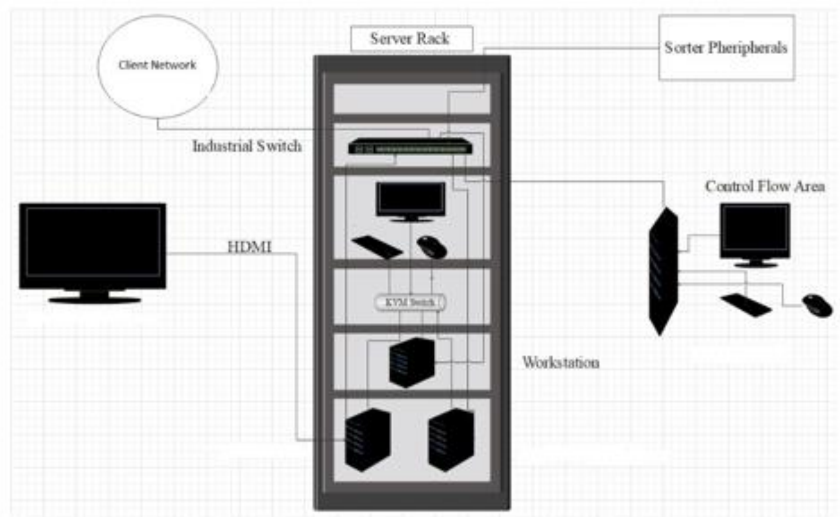
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

11. Controls Architecture (Ref.)



12. IT Architecture (Ref.)



13. Key Components Make

Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Volume Scanners	SICK
Encoders	SICK/Falcon
Sensors	Sick/Leuze/ P&F
PLC	Omron/Siemens
Control Panels	Rittal/ BCH
Weighing Scales	Bizerba/Mettler/Equivalent
VFDs	Siemens/Lenze
Cables	LAPP Equivalent
Switch Gear	Schneider/Equivalent
Bearings	NTN/SKF/Equivalent
Data Transmission System	Siemens
Power Transmission Systems	Vahle
HMIs	Omron/Siemens

14. Program Organisation

14.1 Program/ Project Schedule

System will be ready for UAT by 30th Jan 2025 considering the recipt of Purchase Order by 31st August 2024 .

*, *detailed plan will be shared during DAP.*

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with BLUEDART
3. Weekly/Fortnightly Meeting BLUEDART to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

15. BLUEDART'S Responsibility

15.1 BLUEDART Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the parcelsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.

15.2 Responsibilities of BLUEDART During the Tests

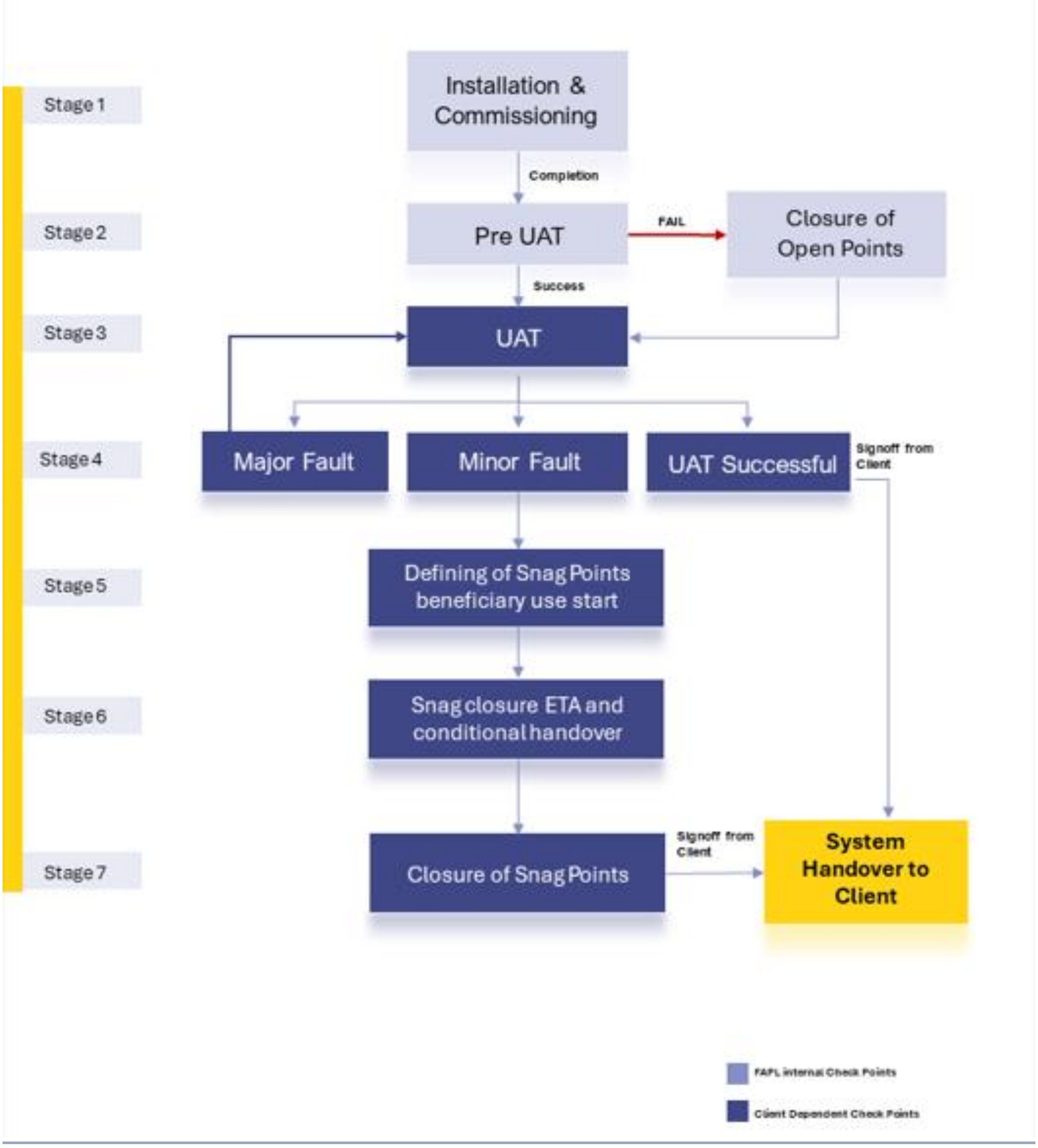
- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the parcels correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

15.3 BLUEDART Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of BLUEDART.

16.System Handover Workflow

The system handover will be following the shown below workflow. Each of the stages are defined in the document below.



Stage 1 - Installation and Commissioning – Completion of all the activities to get the system up and running.

Stage 2 - Pre-UAT – Pre-UAT (User Acceptance Test) involves a series of preparatory steps and tests conducted before the formal UAT phase begins. This stage ensures that the system is ready for end-user testing and meets the necessary requirements and standards. Below are some key checks of Pre UAT:

- **Physical Verification:** Ensure all functional and non-functional requirements are clearly defined and understood. Verify that all the necessary specifications for the sorting system are documented.
- **Integration Test:** Conduct integration tests to ensure all components (conveyors, scanners, sorters, control systems, etc.) work together seamlessly. Validate the interfaces between the sorting system and other warehouse management systems (WMS) and databases.
- **Performance Test:** Test the system's performance with dummy product to ensure it can handle the expected volume of parcels or items. This test also done to identify and rectify any performance bottlenecks or inefficiencies.
- **Functional Test:** Execute test cases to verify that the system performs all required functions correctly. Include scenarios for different types of products, error handling, and edge cases.

By thoroughly conducting pre-UAT activities, Falcon team will ensure that the system is stable, efficient, and ready for the formal UAT phase, where end-users can validate its functionality and performance in a real-world scenario.

Stage 3 – UAT (User Acceptance Test)

3.1 User Acceptance Tests

Based on the Falcon standard User Acceptance Tests parameters (ref Section 3.2 – User Acceptance Tests parameters), Falcon will conduct the User Acceptance Tests (UAT) along with the client. Falcon will provide the UAT plan to client prior to the commencement of any tests. The UAT plan will include the following:

- Test prerequisites
- Daily plan/schedule
- Personnel responsibilities
- Dashboard data and test loads required.
- Specific test procedures
- Expected test results.

3.2 Standard User Acceptance Test Parameters

S. No	Parameter	Value	SOP Document
1	Sorter Throughput	8000 PPH	SOP will be shared during DAP
2	Feedline Throughput	2000 PPH	SOP will be shared during DAP
3	Sorting Accuracy	99.90%	SOP will be shared during DAP
4	Weighing accuracy	+/- 60 grams or 0.05% whichever is higher	SOP will be shared during DAP
5	Dimension Accuracy	+/- 10 mm	SOP will be shared during DAP

Stage 4 – Minor and Major Faults

Any faults encountered during UAT will be categorized by Falcon as either Minor or Major faults as defined below.

- **Minor Faults** are defined as those affecting a limited area or set of components which do not have a functional impact on the system. Minor Faults will be added to the System Snag point list and will not inhibit the continuation of testing.

- **Major Faults** are defined as those having impact that result in the inability to demonstrate the system functionality. Major Faults may require re-starting of the test.

Stage 5 - System Snag Point and Beneficiary use start

System Snag Point list information will include the following:

1. Issue.
2. Date identified.
3. Area and/or unit.
4. Category (mechanical, electrical, controls or software).
5. Remedy responsibility (responsible person[s] or organization[s]).
6. Target completion date.
7. Snag point completion verification and signoff.

In the event that, the UAT is complete with only minor snags, the system would be considered to be in beneficiary use and thus the warranty of the system would be initiated

Note: In the event that Falcon team has called for UAT, and the same is not done by client for 10 days due to reasons beyond the control of Falcon team (e.g., non-availability of dummy or live parcels, manpower or suitable electrical power etc.), the warranty period of the system would be deemed to be initiated

Stage 6 – System Snag Closure procedure

1. Falcon personnel will maintain and distribute the System Sang List throughout UAT.

2. As each issue is corrected, Client will verify the resolution and provide sign-off for that issue on the System Snag Point List.

Stage 7 - System Handover

After successful completion of Acceptance Testing, Falcon will furnish Client a letter, which will address the following:

- The subject System has been installed and accepted by Client.
  NOTE: Acceptance may be conditional and based on agreed dates for snag point closure between Falcon and Client
- The final payment amount that is required and the designated due date.

17. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (**Starts from the date of beneficiary use)** and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Faulty articles continued:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.
  - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
  - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

18. Commercial Terms

18.1 Price Sheet

S.No	Component	Qty per set	Set	UOM	Total (INR)
1	Loop Cross Belt Sorter with VMS, ICR	1	1	Set	₹ 4,69,81,155
2	Arm VDS System	1	1	Set	₹ 30,77,200
3	Conveyors Package	1	1	Set	₹ 88,77,750
4	Chute Package	1	1	Set	₹ 1,29,87,020
5	PTL Frames & Module Package	1	1	Set	₹ 45,56,240
6	Mezzanine & Staircase	1	1	Set	₹ 48,36,852
7	Software Package & Tool Kit	1	1	Set	₹ 2,79,800
8	Installation & Commissioning	1	1	Set	₹ 32,55,841
9	Packaging & Forwarding	1	1	Set	₹ 16,27,920
10	Project Management and Engineering Charges	1	1	Set	₹ 8,64,798
Total					**₹ 8,73,44,576**

Optional Items					
S.No	**Component**	**Qty per set**	**Set**	**UOM**	**Total (INR)**
1	Freight	1	1	Set	₹ 7,20,000
2	DLIC Package Per Year	1	1	Set	₹ 3,00,000
3	Spring Loaded Collection Bin	10	1	Set	₹ 4,00,000
4	Critical Spare	1	1	Set	₹ 48,95,761
5	OSS	1	1	Set	₹ 27,00,000

CMC Details		
Sr No	**Year on Year break-up for CMC charges**	Price
1	1st year after warranty	₹ 44,87,781
2	2nd year after warranty	₹ 44,87,781
3	3rd year after warranty	₹ 44,87,781
4	4th year after warranty	₹ 48,95,761
5	5th year after warranty	₹ 53,03,741
	Total Cost for 5-year CMC per machine (after warranty period) without GST	**₹ 2,36,62,845**

The Total Price is to be understood.

- Taxes: Extra as Applicable
- Including Packing and Forwarding
- Price is valid for 30 days from the date of proposal.

*Material Laydown area is in customer scope

18.2 Payment Terms

S.No	Percentage	Stage
1	40%	Along with Purchase Order within 7 days of contract signing
2	50%	Against Delivery of material at site with 100% Taxes on Pro Data Basis
3	10%	Against Installation & Commissioning

**All invoices (if due as per payment terms) should be cleared within 30 days from the date of receipt of undisputed invoice by the buyer*

19. Comprehensive Maintenance Contract

19.1 Scope of Service

The following services would be covered under this contract:

- E-mail support, Telephonic support, Remote assistance via Team Viewer will be extended to the customer for all issues.
- Preventive Maintenance of the System as per FAPL schedule.
- On-site breakdown service visits & Regular software updates.
- Software related small change requests such as change in the report format, User Interface related changes such as font size, specific color etc.
- Supply, repair/replacement of faulty spare*.
- Supply, replacement of consumables.
- One day workshop for operator training will be provided twice during this service contract.

Sr. No.	Services	Comprehensive Annual Maintenance Package (CMC)
1	Remote Support	✓
2	Break Down Service Visit	✓
3	Preventive Maintenance Visit	✓
4	Software Updates	✓
5	Small Change Request	✓
6	Consumables	✓
7	Spares Warranty	✓
8	Operator Training	2 x one day workshop for operators at client site

**Replacement of spare is subject to failure analysis and approval by FAPL.*

19.2 Requesting Service

- Email: support@falconautoonline.com
- Support Portal: https://falconautotech-pvt-ltd.my.site.com/
- Working Hours: 24 x 7

19.3 Service Response

The following will be the service response for all issues reported to FAPL.

19.3.1 Telephonic and Remote support

Response Time	Severity-0: 60 min Severity-1: 60 min Severity-2: 120 min Severity-3: 120 min

19.3.2 On site physical engineering visit, in case if required.

Engineer Visit Tier-1 City	Severity-0: 24 Hour Severity-1: 24 Hour Severity-2: 36 Hour Severity-3: 48 Hour
Engineer Visit Tier-2 City	Severity-0: 36 Hour Severity-1: 48 Hour Severity-2: 48 Hour Severity-3: 72 Hour

19.3.3 Some Examples of Severity

Severity 0 - System Down	1. Entire loop is down 2. All feedlines are down 3. Server down
Severity 1 - Critical Impact	1. One Zone operation completely down 2. Multiple feedlines affected 3. VDS completely down
Severity 2 - Moderate Impact	1. Any one Feedline affected 2. Bagging conveyor belt damage 3. Any infeed conveyor down

Severity 3 - Low Impact	1. One or more PTL not working 2. One or few carriers not working 3. Slight increase in rejection

19.3.4 Service Level Agreement (SLA) for Spares and Consumables dispatch

Spares & Consumables dispatch	Severity-0: 24 Hour * Severity-1: 24 Hour * Severity-2: 48 Hour * Severity-3: 48 Hour * *If stock available in Plant (Monday 9:00 AM to Saturday 6:00 PM) Excluding Public Holidays

***If stock is not available in plant OEM lead time will be followed.*

19.4 Spares and Consumables transportation responsibility

- The transportation cost of spares & consumables will be in **FAPL scope**.
- The transportation cost of faulty/damaged spares back to Falcon will be in **customer scope**.

19.5 Escalation matrix

- In case the Service request need to be escalated, the following escalation stream should be followed:

Escalation Matrix			
Level	**Name**	**Email**	**Contacts**
1	Nitin – South Zone	nitin.Srivastava@falconautoonline.com	7812997692
2	Sandeep Sharma for sorters	sandeep@falconautoonline.com	85888 80764
	Manish Kumar for non-sorter products	manish.kumar@falconautoonline.com	8447731157
3	Manuj Bansal	manuj@falconautoonline.com	99100 85558
4	Ankit Jain	ankit.jain@falconautoonline.com	81300 65528

19.6 General Terms and conditions with Customer Responsibilities

1. The system entering into the contract should be in good working condition.
2. A health check-up of the system will be done by FAPL on a paid basis before entering into the contract.
3. Proper reporting of the Issue faced via the contact routes specified.
    a. For this it is recommended that customer should train and authorize their employee(s) to coordinate with FAPL (SELLER)
    b. Trained customer employee(s) shall be available on site during each intervention (remote and on site); this(these) employee(s) shall also apply safety and rescue measures in case of emergency.
4. Data transmission requirements for telephonic and remote support (data stream, internet connection speed etc.) – the customer shall ensure the sufficient quality of the telephone and remote connection; any delays and additional costs caused by insufficient data transmission or lack of cooperation from the on-site personnel shall be borne by the customer.
5. The availability of all parts which require customer-specific programming and configuration, within the agreed reaction times for on-site interventions is not guaranteed; the customer shall ensure the on-site availability of these parts in order to avoid delays for the on-site interventions.
6. To send any faulty/damaged parts with proper packaging and documentation back to FAPL, Greater Noida-India, facility for RCA, whenever required.
7. Parts purchased from FAPL must be used and maintained by the customer.
8. In no event shall SELLER be liable to BUYER for any consequential, indirect, special, or punitive damages, including but not limited to loss of business, goodwill, income, profit or savings as well as any other claim for consequential compensation arising out of or relating to this service contract.
9. The BUYER is not authorized/allowed to replace, adjust, repair or in any way manipulate the automated system and its components unless he is specially trained and authorized to perform the corresponding manipulation by FAPL.

19.7 Exclusions

- New IT development.
- Any damage to the equipment due factors such as wrong operation, negligence, improper use, physical damage, damage due to electrical fluctuations/surges/ etc., damage due to Fire, Act of God or any other damage that cannot be attributed to the OEM/FAPL. In such cases, FAPL will take all reasonable steps to ascertain the RCA in coordination with the OEM and the final decision of OEM shall be binding on buyer.
- Training, Data Backup, OS Support, Application Support and Virus Support.
- Dismantling & re-installation of machine related activities.
- Modification/upgradation of the existing setup.

20. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Construction Power**
- **Steel Works- If not specified**
- Building infrastructure;flooring, building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices.
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.